

Section B — Practical Application

Task 1: Workstation Assembly :-

Ans.:-

To assemble a computer workstation by correctly installing the major hardware components and ensuring that the system starts and operates properly.

Steps for Workstation Assembly:-

Step 1: Prepare the Cabinet

- Place the cabinet on a clean, flat surface.
- Remove the side panels.
- Make sure all required screws and mounting points are available.

Step 2: Install the CPU

- Place the motherboard on an anti-static surface.
- Open the CPU socket.
- Align the CPU's marker/notch with the socket.
- Carefully place the CPU into the socket.
- Lock the CPU socket.

Step 3: Install CPU Cooler

- Apply a small amount of thermal paste if required.
- Place the CPU cooler over the processor.
- Secure it properly.
- Connect the cooler's fan cable to the CPU_FAN connector on the motherboard.

Step 4: Install RAM

- Open the RAM slot clips.
- Align the notch on the RAM module with the slot.
- Press the RAM firmly until the clips lock.
- If using two RAM sticks, use the recommended slots mentioned in the motherboard manual.

Step 5: Install the Motherboard

- Install the I/O shield if the motherboard uses a separate one.
- Place the motherboard inside the cabinet.
- Align it with the standoffs and screw holes.
- Secure it using the appropriate screws.

Step 6: Install SSD/HDD

- Mount the SSD/HDD in its appropriate bay or slot.
- For a SATA drive, connect:
 - SATA data cable in motherboard
 - SATA power cable in SMPS
- For an M.2 SSD, insert it into the M.2 slot and secure it with the screw.

Step 7: Install the SMPS

- Place the SMPS in its mounting position.
- Secure it with screws.
- Connect the required power cables to:
 - Motherboard (24-pin ATX)
 - CPU (4/8-pin CPU power)
 - SSD/HDD
 - GPU, if installed

Step 8: Connect Cabinet Front-Panel Cables

Connect the cabinet cables to the motherboard:

- Power SW
- Reset SW
- Power LED
- HDD LED
- USB
- Front audio

Step 9: Connect Monitor and Peripherals

- Connect the monitor to the appropriate video port.
- Connect the keyboard and mouse.
- Connect the power cables.

Step 10: Final Inspection

Check that:

- RAM is properly locked.
- CPU cooler is connected.
- 24-pin and CPU power cables are connected.
- SSD/HDD cables are connected.
- No loose screws or cables are inside the cabinet.
- All components are properly seated.

Step 11: Power On and Test

- Connect the workstation to power.
- Turn on the SMPS.
- Press the cabinet power button.
- Check whether the monitor displays the POST/BIOS screen.
- Enter BIOS/UEFI and verify that the CPU, RAM, and storage device are detected.

Task 2: RAM Mismatch Troubleshooting :-

Ans.:-

Troubleshoot and resolve a RAM mismatch error in a startup office workstation using BIOS/UEFI settings. The aim is to identify the error through POST messages, determine its root cause, correct the RAM configuration, and confirm that the workstation boots normally.

Troubleshooting Procedure:-

Step 1: Start the Workstation

- Turn on the office workstation.
- Observe the POST (Power-On Self-Test) screen.
- Enter BIOS/UEFI by pressing the required key such as Delete, F2, or F10.

Step 2: Identify the RAM Error

The BIOS/UEFI showed a memory-related warning such as:

Memory Configuration Mismatch /RAM Speed Mismatch / Memory Size Mismatch

The BIOS displayed the installed RAM modules and their specifications.

Step 3: Check RAM Specifications :-

The root cause was identified as a RAM speed/timing mismatch. The two RAM modules were compatible in generation but had different rated speeds and timings.

Step 4: Correct the Configuration:-

There are two possible solutions:

Option 1 – Reconfigure BIOS/UEFI:-

- Open the BIOS/UEFI memory settings.
- Disable an incompatible memory overclock/XMP setting if necessary.
- Set the RAM to a common supported speed, for example 2666 MHz.
- Save the settings and restart the computer.

Option 2 – Reseat/Replace RAM:-

- Shut down the workstation.
- Disconnect the power cable.
- Remove the RAM modules.
- Check that there is no dust or damage.
- Reinstall the RAM firmly into the recommended motherboard slots.
- If necessary, replace the mismatched module with a matching RAM module.

Step 5: Verify BIOS/UEFI

After restarting:

- Enter BIOS/UEFI again.
- Check the total memory capacity.

- Confirm that both RAM modules are detected.
- Verify that they are operating at the selected compatible speed.

For example:

Total Memory: 16 GB

Memory Speed: 2666 MHz

Status: Both RAM modules detected

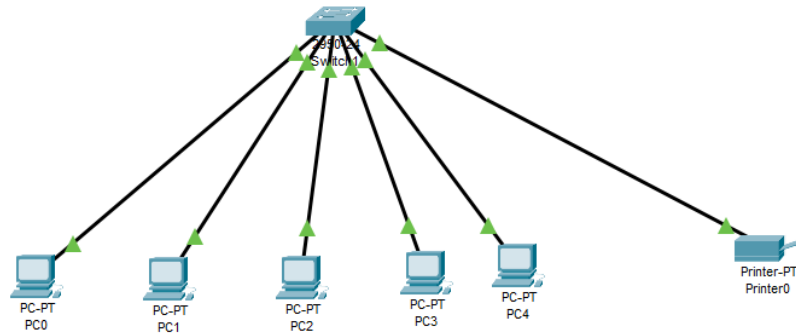
Step 6: Confirm Normal Boot:-

- Save BIOS/UEFI settings.
- Restart the workstation.
- Confirm that the RAM mismatch warning no longer appears.
- Allow Windows to start normally.
- Check the installed RAM from Task Manager in Performance in Memory.

The RAM mismatch error was successfully identified through the BIOS/UEFI POST message. The root cause was a difference in RAM speed/timing. The RAM was correctly reseated and configured to a compatible speed. After saving the BIOS/UEFI settings, the workstation booted normally and detected the full 16 GB of RAM.

Task 3: LAN Topology Build:-

Ans.:-



Device	Ip address	Subnet mask
Pc0	192.168.1.1	255.255.255.0
Pc1	192.168.1.2	255.255.255.0
Pc2	192.168.1.3	255.255.255.0
Pc3	192.168.1.4	255.255.255.0
Pc4	192.168.1.5	255.255.255.0
Printer	192.168.1.6	255.255.255.0

Network: 192.168.1.0/24

Subnet Mask: 255.255.255.0

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time=2ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Task 4: Clean OS Installation :-

Ans.:-

To perform a clean installation of an operating system from an ISO file on an office workstation, manually partition the disk, and optimize startup programs/services to improve boot performance.

Step 1: Backup Important Data :-

Before starting the installation:

- Back up important documents, pictures, and office files.
- Save data to an external drive or another safe location.

Step 2: Create a Bootable USB :-

Using another computer:

1. Connect a USB flash drive.
2. Use an appropriate ISO-to-USB tool.
3. Select the Windows/Linux ISO.
4. Select the USB drive.
5. Create the bootable USB.

Step 3: Boot From the ISO/USB :-

1. Insert the bootable USB into the workstation.
2. Power on the computer.
3. Open the Boot Menu or BIOS/UEFI.
4. Select the USB device as the boot device.
5. Start the operating-system installer.

Step 4: Start the Clean Installation :-

For Windows:

1. Select language.
2. Select time/currency format.
3. Select keyboard layout.
4. Click Install Now.
5. Select the required Windows edition.
6. Accept the license terms.
7. Select Custom: Install Windows only.

Step 5: Manually Partition the Disk :-

Suppose the workstation has a 256 GB SSD.

1. Select the target disk.
2. Delete old partitions only after confirming the backup.

3. Create the required system partitions.
4. Create the OS partition.
5. Create a separate data partition if required.
6. Select the OS partition.
7. Click Next.

Step 6: Install the Operating System

The installer will:

- Copy installation files.
- Install system components.
- Configure the bootloader.
- Install drivers.
- Restart the workstation several times.

Step 7: Install Drivers :-

Check Device Manager.

Verify that there are no devices showing a warning symbol.

Step 8: Optimize Startup Programs :-

Open:

Task Manager in Startup apps

Review the programs that start automatically.

Disable unnecessary programs such as:

- Unused launchers
- Unnecessary update assistants
- Applications that are not required at startup

Step 9: Optimize Startup Services:-

Win + R click and types services.msc and enter.

Only disable services that are confirmed to be unnecessary for the workstation and its required applications.

This can reduce unnecessary background activity during startup.

Step 10: Check Boot Configuration :-

System Configuration in Boot

Verify that the correct operating system is selected.

For a normal modern Windows installation, the workstation should use UEFI where supported.

Step 11: Test the Installation :-

Restart the workstation several times.

Check that:

- Windows/Linux starts without errors.
- No boot-loop occurs.
- The correct OS partition is used.

- Keyboard and mouse work.
- Network connectivity works.
- Audio and display work.
- All required drivers are installed.

A clean operating-system installation was successfully performed from an ISO using a bootable USB.